

20. StockSense — Stock Control POS & Store Remodeling Recommender

Domain	Web + Mobile — POS systems + data analytics
Team size	3 students
The problem	Small retail stores track stock manually and have no data-driven sense of store layout — which items should be placed together or in high-traffic zones based on what actually sells and sells together.
Users	Store staff at checkout; store manager reviewing stock and layout decisions.

Requirements

POS & inventory

- **MUST** A working point-of-sale flow: staff select items, set quantities, complete a sale, and stock is decremented correctly — including under near-simultaneous sales from two terminals on the last unit of an item (a real concurrency case, not just a happy path).
- **MUST** Manager can add/edit stock items and view current levels, with a configurable low-stock alert threshold.
- **SHOULD** Quick manual SKU/code entry to speed up checkout (doesn't require real barcode-scanner hardware).

Reporting

- **MUST** Sales report by item/category over a selectable date range.
- **MUST** Identify best-sellers and slow-movers over a period from actual recorded sales data, not a guess.

Store remodeling recommendation

- **MUST** Using recorded sales data (item frequency and co-purchase patterns), generate a store-layout recommendation — e.g. group frequently co-purchased items, place best-sellers in high-traffic zones. Simple co-occurrence counting or association-rule mining (e.g. Apriori) is appropriate; no exotic ML required.
- **SHOULD** Visualize the recommendation as a simple layout/floor-plan diagram showing suggested zones, not just a text list.
- **COULD** Let the manager simulate a proposed layout change and see an estimated improvement metric (e.g. reduced average distance between commonly co-purchased items).

Mobile companion

- **MUST** A mobile app for the manager to check stock levels, alerts, and reports on the go — read-focused, doesn't need full POS functionality.

Out of Scope

Computer-vision-based shelf/stock detection and any in-store robotic or AR guidance — the layout recommendation is a data/analytics deliverable, not a hardware project.

Suggested Technologies

Layer	Suggested technologies
POS/staff frontend	React (web) for the till/checkout interface
Manager mobile app	React Native or Flutter
Backend	Python FastAPI or Node.js/Express

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Database	PostgreSQL, modeled carefully for transactional stock decrements
Analytics	pandas for co-occurrence analysis; mlxtend (Apriori) for association-rule mining
Layout visualization	A simple grid-based SVG floor plan, or Recharts/D3 for the recommendation view
Real-time sync (optional)	WebSockets to keep multiple POS terminals' stock counts consistent live
Hosting/Deploy	Render or Railway for backend/DB; Expo for the mobile app

Definition of Done

Complete a few sale transactions live on the POS; show stock decrementing correctly, including one blocked sale when two terminals compete for the last unit; pull a sales report; and show the generated layout recommendation with its reasoning traced back to actual co-purchase data from seeded transactions.